EDITORIAL

Low compliance with the United States Multi-Society Task Force on Colorectal Cancer postpolypectomy guidelines

Screening colonoscopy prevents colorectal cancer (CRC) through the detection and resection of precursor lesions. The goal of the index colonoscopy is not only to clear the colon of these precursors but to risk stratify patients for surveillance based on guideline recommendations. Although surveillance colonoscopies reduce the risk for postcolonoscopy CRC by detecting missed, incompletely resected, and incident polyps or CRC, these examinations have attendant risks and costs.[1] Guidelines are crafted to balance the risks and benefits of surveillance. Thus, compliance with these guidelines is important for effective CRC screening and surveillance.

The United States Multi-Society Task Force on Colorectal Cancer (USMSTF) published updated guidelines in 2020 on colonoscopy surveillance intervals.[1] The most notable guideline change was the increase in the interval for 1 or 2 small (<1cm) tubular adenomas, known as low-risk adenomas (LRAs), from a surveillance interval of 5 to 10 years to 7 to 10 years. Other changes include further guidance for sessile serrated polyps (SSPs) and large (≥1 cm) hyperplastic polyps (HPs), and lengthening the surveillance interval for 3 or 4 small tubular adenomas to 3 to 5 years.

In this issue of *Gastrointestinal Endoscopy,* Dong et al[2] publish their assessment of the compliance of 33 endoscopists at a single center with the updated 2020 USMSTF polypectomy surveillance guidelines. The authors analyzed data from screening colonoscopies performed before and after the 2020 guideline change. They observed that adherence was 48.9% for all polyps, 8.3% for LRAs, 88.3% for high-risk adenomas, 63.1% for SSPs, and 88.6% for HPs. When these data were reassessed for compliance according to the older 2012 surveillance guidelines, adherence rates improved dramatically, for LRAs in particular. These data suggest that endoscopists may be slow to adopt the new recommendations for surveillance intervals for LRAs.

This novel study has many strengths, including the use of first-time screening colonoscopies only, excluding surveillance examinations in which follow-up intervals may be based on previous neoplasia. They also examined adherence for different index findings over a few distinct time periods after the guidelines were published, and they observed no change in adherence. The use of a survey for endoscopists helped to demonstrate gaps of knowledge for surveillance recommendations. Whereas adherence for LRA surveillance was higher in those who identified the correct interval for LRAs, the adherence was still unacceptably low. The investigators also identified factors associated with noncompliance, including endoscopists who had finished their training ≥10 years earlier and those who performed >800 colonoscopies per year.

The limitations of this study are mostly due to a small sample size (33 endoscopists) at a single institution, which may make the results less generalizable. The survey also had a response rate of only 57% (19 endoscopists). The postguideline time intervals may also have been too short to enable observation of significant improvement, and future studies should examine compliance over a longer time. In addition, only gastroenterologists were included in the analysis; thus, the authors could not examine compliance by other specialists.

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Whereas the main findings of this study demonstrate low compliance among endoscopists for surveillance recommendations for LRAs, the survey results suggest that the low adherence may not be explained simply by a lack of guideline awareness. Specifically, endoscopists with correct answers to LRA clinical vignettes still demonstrated nonadherence in their clinical practice. Previous studies have demonstrated similar noncompliance with LRA surveillance. In 1 study, factors associated with shorter surveillance intervals for LRA included patient's family history of CRC, quality of bowel preparation, and number and size of polyps.[3] In the current analysis, patients with a family history of CRC, genetic syndromes, and incomplete examinations or poor bowel preparation were excluded; therefore; the impact of these factors could not be examined.

It is possible that the reluctance among gastroenterologists to change their practice patterns may be due to data suggesting that the risk of metachronous neoplasia may be

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TABLE 1. USMSTF postpolypectomy guidelines and compliance tips for select groups

Lesion	Findings	Change (years) from 2012	Rationale	Tips for managing/compliance
Adenomas	Adenoma ≥1 cm, with villous elements and/or high-grade dysplasia	Same: 3	• Elevated risk for incident CRC	• Maintain adequate ADR • Quality examination; 8-9 min withdrawal time • Complete resection, snare (cold preferable) • Forceps for <3 mm only if snare not feasible (ie, polyp in left upper part of lumen) • Communicate with patient about interval • Educate referring physicians/providers about surveillance • Can use longer interval for concern about risk (eg, 7 vs 10 years for 8-mm adenoma)
	5-10 small (<1 cm) TAs	Same: 3	• Elevated risk for incident CRC	
	3-4 small (<1 cm) TAs	3 to 3-5	• Better technology and attention to quality has increased yield • Low risk for metachronous advanced neoplasia • Low risk for incident CRC	
	1-2 small (<1 cm) TAs	5-10 to 7-10	• Similar risk for incident CRC to normal examination • Better technology and attention to quality has increased yield	
Serrated polyps	Large SSP or SSP w/dysplasia or TSA	Same: 3	• Increased risk for incident CRC in long-term studies	• Maintain adequate serrated detection rate • Quality examination; 8-9 min withdrawal time • Communicate with pathologist about SSP vs HP • Careful resection with EMR for larger serrated polyps • Be cognizant of serrated polyposis syndrome (cumulative diagnosis)
	HPs ≥1 cm	10 to 3-5	• Increased risk for incident CRC in long-term studies	
	5-10 small (<1 cm) SSPs	5 to 3	• Increased risk of metachronous large SP	
	3-4 small SSPs	5 to 3-5	• Increased risk of metachronous large SP	
	1-2 small (<1 cm) SSPs	5 to 5-10	• Increased risk of metachronous large SP	
	< 20 HPs proximal to sigmoid colon (<1 cm)	10	• Low risk for metachronous large SPs	

ADR, Adenoma detection rate; *CRC*, colorectal cancer; *HP*, hyperplastic polyp; *SP*, serrated polyp; *SSP*, sessile serrated polyp; *USMSTF*, United States Multi-Society Task Force.

different depending on the size and number of LRAs. The current guidelines do not distinguish between small (5- to 9-mm) versus diminutive (<5-mm) polyps. However, published data suggest that the presence of at least 1 small (5- to 9-mm) adenoma in patients with LRAs may be associated with an increased metachronous risk of advanced adenomas as compared with index findings in which both adenomas were diminutive.[4,5] Thus, potential heterogeneity for future neoplasia risk may lead to uncertainty and lack of confidence in the recommendations. Specifically, endoscopists may not be as comfortable with a longer interval for 9-mm adenomas as they may be with diminutive adenomas.

Although short-term data examining risk for metachronous advanced neoplasia suggest a heterogeneity of risk for LRAs, data for long-term risk for CRC are imperfect and may not provide clear answers as well. Prior published data have suggested that patients with LRAs, even those with multiple small adenomas (≥3), may have a long-term risk for incident CRC, which is similar to those with normal examinations.[6] However, these data have wide confidence intervals, which precludes any definitive conclusions about their risk.[6] In addition, the low-risk groups were more likely than the normal index examination group to have had surveillance colonoscopies, which may have mitigated the incident CRC risk in the low-risk group. These imperfect data may also add to uncertainty about the long-term risk for LRA and therefore lead to noncompliance because of reluctance to recommend longer intervals.

The authors also observed that guideline adherence for SSPs was equally low in both the pre- and postguideline periods. There is heightened awareness for serrated polyps among endoscopists because the serrated pathway may be responsible for ≤20% to 30% of all CRCs.[7,8] Furthermore, there may be uncertainty among endoscopists with respect to management of serrated polyps for many reasons. Inasmuch as SPs are often flat with indistinct borders, they can be difficult to detect, as manifested by the wide variation in endoscopist detection rates. Their flat appearance is likely the reason for the high incomplete resection rates for SSPs, which approaches nearly 50% for large serrated polyps.[9] Incompletely resected polyps may play a critical role in postcolonoscopy CRC.[9,10] Thus, endoscopists may recommend a shorter surveillance interval for SSPs if they lack confidence that they detected all of the SSPs or that they have achieved complete resection.

Distinguishing SSPs from HPs can also be difficult endoscopically as well as histologically.[11] The clinical implications for these serrated polyps are different, given that small and distal HPs are believed to have a benign natural course whereas SSPs are believed to have a potential for malignant transformation. Because some HPs, such as those 5 to 9 mm and proximal, may be misdiagnosed SSPs, it is possible that endoscopists may suggest shorter guidelines than the 10-year intervals recommended for HPs <1 cm. In addition, some data suggest that 5- to 9-mm HPs have a significant increased risk for future large SPs.[12] However, in the current study the compliance for all HPs was close to 90%. In terms of subsets of small (5-9 mm) proximal HPs, there were too few patients with these findings to enable conclusions to be drawn.

What are our recommendations for gastroenterologists and other endoscopists? It is important that endoscopists update their knowledge base regarding surveillance guidelines (Table 1). In addition, they need to communicate effectively with referring physicians and providers. It is also important to discuss the recommendations with patients because some might expect shorter intervals based on their experience or previous endoscopists.

Endoscopists can increase their confidence with the longer surveillance intervals that are currently recommended by performing high-quality colonoscopies and ensuring that the index colonoscopy successfully clears the colon of all adenomas and serrated polyps. An important recommendation of the USMSTF postpolypectomy guidelines is that the index examination should be a high-quality colonoscopy. Although the authors of the current study excluded inadequate bowel preparation and incomplete examinations, they did not provide data on endoscopist detection rates. Recent data from 2 screening cohorts demonstrate that the incidence of CRC after polypectomy for low-performing endoscopists was twice that for high-performing endoscopists in patients with LRAs.[13]

High-quality colonoscopy can be accomplished by adhering to quality metrics that have been shown to increase adenoma detection rates, such as a withdrawal time of $\geq$9 minutes,[14] and by ensuring that the patient has had an optimal preparation.[15] For gastroenterologists and other endoscopists to have increased confidence in their ability to clear the colon, it is also important to have confidence in polypectomy techniques. By following current polypectomy technique recommendations, such as cold snaring (superior to forceps) and use of EMR for large polyps, endoscopists can increase their confidence in clearing the colon of adenomatous and serrated polyps.[16]

The recommended screening intervals for LRA are likely to be extended. Emerging long-term CRC data support a 10-year surveillance interval for LRAs.[6] But more data, especially those from randomized trials, are needed to identify the best interval for LRAs. The results from FORTE (Five or Ten Year Colonoscopy for 1-2 Non-Advanced Adenomatous Polyps; NCT05080673) are expected in approximately 10 years. Current guidelines from the United Kingdom and the European Society of Gastrointestinal Endoscopy recommend longer intervals of 10 years for patients who have <5 small (<1 cm) tubular adenomas on index colonoscopy.[17,18] Thus, it is important that endoscopists keep current and comply with new surveillance intervals, especially those for the LRAs. By completing high-quality colonoscopies and polypectomies, gastroenterologists can be confident that the risk of postcolonoscopy cancer is significantly reduced, and they can have the confidence to adhere to the 2020 USMSTF postpolypectomy surveillance guidelines.

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Abbreviations: CRC, colorectal cancer; HP, hyperplastic polyp; LRA, low-risk adenoma; SP, serrated polyp; SSP, sessile serrated polyp; USMSTF, United States Multi-Society Task Force.

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